SHORT UGC VIDEO QUALITY ASSESSMENT VIA LARGE LANGUAGE MODEL: BENCHMARK AND METHOD

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ABSTRACT

Video quality assessment (VQA) is a core component of modern video systems which can guide adaptive delivery and supporting A/B testing. Recent advances in multimodal large language models (MLLMs) extend VQA beyond hand-crafted approaches and conventional neural regressors through pretrained semantic knowledge and language-based quality prediction. However, most existing MLLM-based VQA methods are context-agnostic, assigning a single quality score to each video. This formulation does not directly accommodate the diverse assessment requirements of video systems. In this paper, we consider *transcoding* as a representative task, where perceived quality depends on both the source video and the viewing conditions. To address this limitation, we first construct TSVQ, a subjective benchmark for evaluating short-form video transcodes against their uploaded sources under diverse mobile viewing conditions. Our analysis reveals divergent human and MLLM responses to motion under heavy compression, together with differences in quality ratings associated with source and screen resolution. Guided by these findings, we propose MOCA-VQA, which has two components, *Motion-aware Quality Scorer* (MQS) and *Context-aware Score Rectifier* (CSR). MQS wisely combines spatial artifact cues with aligned motion features to model the perceptual degradation of transcoded videos. CSR then adjusts the score using content features and context features describing the source, encoding configuration, and viewing screen. Experiments show that MQS achieves statistically significant gains over the evaluated baselines on TSVQ and open-source dataset, while CSR improves average screen-tier performance across multiple base scorers.

1 INTRODUCTION

Short user-generated video (UGC) has become a dominant form of online entertainment, consumed and shared by billions of users daily on social media platforms such as TikTok, YouTube Shorts, and Instagram Reels (TikTok, 2021; Mohan, 2024). To serve billions of views a day, platforms must take in uploaded videos of highly uncontrolled quality and deliver them over heterogeneous networks to various devices while keeping the viewing experience consistent. Accurate and reliable video quality assessment (VQA) is therefore a core component of short-video delivery platforms, as it enables quality-of-experience (QoE) monitoring at scale, guides transcoding and adaptive delivery, and supports A/B testing (Dobrian et al., 2011; Krishnan & Sitaraman, 2012; Li et al., 2016; Wang et al., 2021).

Owing to its central role in video systems, VQA has been extensively studied in both academia and industry, evolving from hand-crafted models built on natural scene statistics (Mittal et al., 2012; 2013; Saad et al., 2014), through end-to-end convolutional and transformer-based regressors (Sun et al., 2022; Lu et al., 2024; Wu et al., 2022; 2023a), to multimodal large language models (MLLMs) that encode a video as visual tokens and predict quality in the language space (Wu et al., 2024; You et al., 2025; Wen et al., 2025; Jia et al., 2024; Ge et al., 2024; Li et al., 2025; Zhang et al., 2026). MLLM-based scorer achieves higher prediction accuracy and markedly better cross-content generalization capability than the traditional approaches due to its semantic priors acquired through

large-scale pretraining and a language decoder capable of reasoning about visual content (Wu et al., 2024; Li et al., 2025).

However, most of MLLM-based VQA methods share one formulation: given a single video, output a score reflecting how good it looks in general (Wu et al., 2024; You et al., 2025; Ge et al., 2024; Li et al., 2025; Zhang et al., 2026). Such prediction is inherently *context-agnostic* and does not directly transfer to every task in a delivery pipeline. In this paper, we fcous on *transcoding*, one of the most critical operations in video systems where every uploaded source video is re-encoded into multiple representations to accommodate varying network conditions and is viewed on heterogeneous devices. Here, quality depends not only on the video itself, but also on the context in which it is produced and viewed. The goal is thus to measure *perceptual loss introduced by transcoding relative to the upload on a given device*.

Current MLLM-based scorers fall short of this goal in two aspects. *First of all, the quality degradation itself is largely invisible to these models.* Specifically, they inspect a handful of sparsely sampled and aggressively resized frames for cost efficiency. While well suited to semantic understanding, this abstraction destroys the fine-grained spatiotemporal evidence that determines whether a compression artifact is visible (Wu et al., 2022; Wen et al., 2025). *Second, these models have never been taught to place the perceptual loss in its context.* The datasets they are trained on (Ying et al., 2021; Hosu et al., 2017; Sinno & Bovik, 2019; Wang et al., 2019) provide a single absolute rating per video, collected without any notion of a source version or a viewing condition. A model fit to such labels cannot fully express the perceived loss in real viewing scenarios. Prior work partially addresses these limitations. Specifically, CP-LLM (Wen et al., 2025) sharpens MLLMs' sensitivity to compression artifacts but leaves temporal dynamics unmodeled, and device-aware subjective studies are confined to professionally generated content (Safonov & Vatolin, 2026; Barman et al., 2023; Moorthy et al., 2012). How to adapt MLLM-based VQA methods for perceptual loss evaluation in short-form UGC video transcoding remains unexplored.

In this paper, we bridge this gap through task-specific data and modeling. In particular, we first conduct a large-scale controlled subjective study and build TSVQ (Transcoded Short-form Video Quality). We collect 690 source videos from a leading short-video platform (TikTok), spanning diverse content categories and five native resolutions, and transcode each along a resolution $\times$ CRF $\times$ codec ladder, yielding 12600 pairs of videos and 377K ratings in total. With TSVQ, we systematically study how humans perceive transcoding loss under different viewing contexts and how existing MLLM-based predictors perform on TSVQ. Our key findings are as follows. First, *sparse-frame MLLM scorers misestimate motion-dependent degradation.* Whereas humans tolerate heavily compressed frames in dynamic content owing to motion masking, MLLMs uniformly penalize them. Second, *perceived loss is source- and device-dependent.* Among device attributes, display resolution has the strongest effect, with quality gains saturating once the video resolution reaches that of the display. At matched transcoded resolutions, ratings also differ between source-resolution groups.

Guided by these findings, we design MoCA-VQA, a task-specific MLLM scorer for short-form UGC transcoding. MoCA-VQA consists of two components, a multi-branch MLLM base scorer and a rectifier for context adaptation. First, we provide the MLLM with four aligned views of the video: resized full frames, full-resolution patches, and the motion of each. Artifact and motion cues are fused before entering the decoder, capturing the motion-dependent visibility of compression artifacts without inflating the token budget. Second, we append a rectifier network after the MLLM decoder to capture the score shifts induced by the viewing context. Conditioned on the source resolution and device attributes together with features of the transcoded video and the base score produced by the MLLM, the rectifier outputs the final perceptual quality. Notably, the rectifier requires only the source metadata rather than the source video itself. Such information is routinely logged in production systems, keeping MoCA-VQA no-reference and practical at serving time. Our evaluation shows that MoCA-VQA outperforms the existing MLLM scorers and conventional regressor on TSVQ and public benchmark, paving the way for MLLMs to serve as perceptual measurement infrastructure in video delivery systems.

In summary, our contributions are as follows. (i) We build TSVQ, the first benchmark that jointly controls source resolution, transcoding ladder, and viewing device to measure short-form UGC transcoding loss. (ii) We identify two key systematic gaps between existing MLLM-based scorers and human viewers: they fail to reproduce motion masking and to capture context-dependent

loss. (iii) We propose MoCA-VQA, a context-aware MLLM scorer whose two components each address one gap, achieving state-of-the-art accuracy on TSVQ and public benchmarks.

2 Related Work

2.1 VQA before MLLMs

Early no-reference quality assessment methods mainly rely on handcrafted statistical features (Mittal et al., 2012; 2013; Saad et al., 2014; Mittal et al., 2016). Some hybrid approaches further incorporate pretrained deep features (Tu et al., 2021). Deep learning-based VQA captures spatial distortions and temporal dynamics through learned representations, which are aggregated for quality regression (Ying et al., 2021; Sun et al., 2022; Wu et al., 2022; Wen et al., 2024). Further advances improve quality modeling through specialized architectures (Wu et al., 2023a; Shen et al., 2025; Lu et al., 2024) and adaptation of pretrained visual features to perceptual assessment (Yuan et al., 2024; Li et al., 2024). Conventional VQA methods often exhibit limited cross-dataset generalization, with prediction accuracy dropping on new dataset.

2.2 MLLMs for VQA

Recent work leverages the semantic priors and language modeling capabilities of MLLMs for perceptual quality assessment. Q-Align (Wu et al., 2024) encodes visual inputs into tokens using a vision encoder and predicts quality scores through an LLM decoder adapted via supervised fine-tuning (SFT). DeQA (You et al., 2025) replaces Q-Align's one-hot quality label with a soft probability distribution over quality levels. To capture temporal distortions, LMM-VQA (Ge et al., 2024) and VQA^2 (Jia et al., 2024) augment spatial representations with features extracted by motion encoders. In addition, recent work tailors MLLMs to perceptual quality assessment by combining global context with high-resolution patch features to capture fine-grained distortions (Wen et al., 2025) or extending overall scoring to multidimensional quality assessment and distortion identification (Duan et al., 2025). Another line of research explores reinforcement learning (RL) to elicit quality-aware reasoning from MLLMs while reducing reliance on large-scale and high-quality annotations (Li et al., 2025; Wu et al., 2025; Zhang et al., 2026; Wen et al., 2026).

2.3 VQA datasets

Most existing UGC datasets primarily provide absolute quality ratings for videos for general purposes rather than measurements of transcoding-induced degradation (Ying et al., 2021; Hosu et al., 2017; Sinno & Bovik, 2019; Wang et al., 2019). Transcoding-oriented datasets assess quality difference relative to uploaded sources, i.e., DMOS (Li et al., 2020b; Wang et al., 2021). BVI-UGC (Qi et al., 2024) further evaluates non-pristine UGC across multiple transcoding configurations. However, these datasets either use fixed laboratory displays or rely on crowdsourced evaluations without controlled variation in viewing devices. Other datasets explicitly account for viewing conditions but focus on high-quality reference content rather than transcoding from non-pristine UGC uploads (Duanmu et al., 2020; Safonov & Vatolin, 2026). As summarized in Table 1, TSVQ is designed to assess short-form UGC transcoding quality under controlled realistic mobile viewing conditions.

3 TSVQ: Benchmark and Key Findings

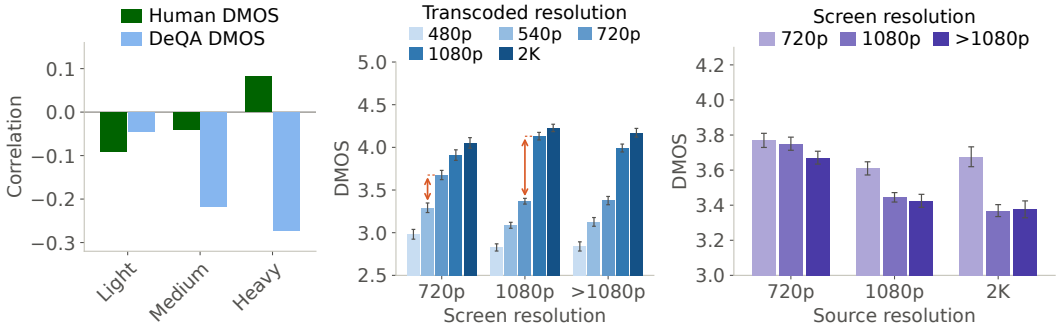
In this section we describe our dataset, TSVQ and the key findings drawn from it. Following the double-stimulus impairment scale (DSIS) method in ITU-R BT.500 (ITU-R, 2019), we collect DMOS for each transcoded video v relative to its original upload u, with the two videos presented side by side on the phone. Details of the subjective study and the complete analysis are given in Appendix **??**.

3.1 Dataset Overview

[[Needs to highlight why this dataset is unique – not just another dataset. Can we list the number of rating tasks in the table to show the dataset's scale?]]

Table 1: Comparison of TSVQ with existing subjective VQA datasets. *DMOS*: the perceptual quality of a transcoded video rated relative to its source video. *Ladder*: the transcoding axes varied per source. *Viewing conditions*: ✓ marks studies in which the viewing device is deliberately controlled and recorded for every rating.

Dataset	Content	#Videos	Rating	Ladder (per source)	Viewing conditions
LSVQ (Ying et al., 2021)	UGC	39,075	absolute	–	×
KoNViD-1k (Hosu et al., 2017)	UGC	1,200	absolute	–	×
LIVE-VQC (Sinno & Bovik, 2019)	UGC	585	absolute	–	×
YouTube-UGC (Wang et al., 2019)	UGC	1,500	absolute	–	×
UGC-VIDEO (Li et al., 2020b)	UGC, short	550	DMOS	codec × QP (10)	×
YT-Compress (Wang et al., 2021)	UGC	189×3	DMOS	bitrate (3)	×
TaoLive (Zhang et al., 2023)	UGC, live	3,762	absolute	CRF (8)	×
KVQ (Lu et al., 2024)	UGC, short	4,200	absolute, ranks	QP, +enhancement (6)	×
MaxWell (Wu et al., 2023b)	UGC	4,543	absolute, 13 dims	–	×
BVI-UGC (Qi et al., 2024)	UGC	1,080	DMOS	res. × codec × QP (18)	×
FineVD (Duan et al., 2025)	UGC	6,104	absolute, 6 dims	–	×
YT-Shorts (Wang et al., 2024)	UGC, short	2,030+2,000	absolute	–	×
CHUG (Saini et al., 2025)	UGC, HDR	5,992	absolute (0–100)	res. × bitrate (6)	×
Beyond8Bits (Saini et al., 2026)	UGC, HDR	44,276	absolute (0–100)	res. × bitrate (6)	×
SQoE-IV (Duanmu et al., 2020)	PGC	1,350	absolute	res. × bitrate × codec	✓
Safonov & Vatolin (2026)	PGC	25×26	pairwise	codec × bitrate/QP (26)	✓
TSVQ (ours)	UGC, short	12,600	DMOS	src res. × res. × CRF × codec	✓



(A) Spearman correlation between content TI and score at three compression levels: human and MLLM score diverges.

(B) DMOS of transcoded videos from 2K sources *vs.* screen resolutions: DMOS gains peak at the screen resolution.

(C) DMOS of 720p transcoded videos *vs.* source resolutions: viewers rate more strictly when source resolution is higher.

Figure 1: Three key findings from TSVQ: fully capturing the perceptual loss introduced by UGC transcoding requires jointly considering content degradation, device context, and reference context.

TSVQ contains 690 source videos selected from approximately 120,000 uploads to a leading short-video platform TikToK. The sources span 22 content genres and five native resolutions: 540p, 720p, 1080p, 2K, and 4K. We select source videos to cover various content diversity and spatial and temporal complexity, measured by spatial information (SI) and temporal information (TI) (ITU-T, 2023; Li et al., 2020a). Each source is then transcoded to target resolutions of 480p, 540p, 720p, 1080p, 2K, and 4K that do not exceed its native resolution at selected CRF values between 22 and 34. Together with different codec, this produces 12,600 transcoded videos. The study uses 25 smart phones with screen widths of 720–1440 px and pixel densities of 269–505 PPI. These devices are grouped into three screen-width tiers, $\mathcal{T} = \{720\text{p}, 1080\text{p}, >1080\text{p}\}$, respectively. Each rater is assigned a phone, and the device model is recorded for every rating. In total, we have 48 raters that contribute 377,050 valid ratings. Further details are provided in Appendix **??**.

3.2 Key Findings

In this part, we examine three key factors that guide the design of our quality predictor: content motion, screen resolution, and source resolution. The complete analysis is given in Appendix **??**.

[[Say something about the novelty of these findings? Were they identified qualitatively or quantitatively previously?]]

Content motion: human and model responses diverge under degradation. Fig. 1A compares the correlation of TI, which measures temporal dynamics, with human DMOS and DeQA predictions (You et al., 2025), one of the MLLM-based VQA approaches, across degradation levels.[1] Both correlations are weak under mild degradation. As degradation becomes more severe, however, DeQA predictions exhibit a stronger negative correlation with TI, whereas human ratings remain largely uncorrelated. One reason is that content dynamics during playback can reduce the visibility of certain distortions (Choi & Bovik, 2018). Sparse frame sampling may inadequately capture this perceptual effect in MLLM-based quality prediction. This divergence motivates jointly modeling artifacts and their motion context to better approximate distortion visibility during playback for MLLM (Sec. 4.1).

Screen resolution: perceptual gains depend on the display. Fig. 1B illustrates the DMOS versus the screen resolution, and we found that the same increase in video resolution yields different perceptual gains across screen-resolution groups. This pattern is particularly evident on 1080p screens, where the largest gain occurs when video resolution reaches the screen resolution, and further increases yield substantially smaller improvements. These findings motivate us to consider screen resolution when predicting the perceptual quality (Sec. 4.2).

Source resolution: quality ratings depend on reference context. Fig. 1C compares DMOS for transcoded videos of 720p derived from various source resolutions. The 1080p-source and 2K-source groups receive lower mean ratings, potentially because viewers perceive greater degradation relative to higher-quality sources. The magnitude of this difference also varies with screen resolution. In particular, the gap is larger on 1080p and ¿1080p screens. These results further suggest that the perceived quality of transcoded videos depends on both reference and device context (Sec. 4.2).

4 METHOD

[Yu#1: overall i think this section is too short.]

In this section, we present MOCA-VQA, which combines an MLLM-based *Motion-aware Quality Scorer* (MQS) with a *Context-aware Score Rectifier* (CSR), as illustrated in Fig. 2. MQS operates in a no-reference setting [[Need to motivate the no-reference setting. In intro, you said you focus on transcoding, which usually has the reference (source), so why adopt no-reference setting here?]] and predicts a base perceptual degradation score $q_0(v)$ from the transcoded video v. CSR then adjusts this score using video content features and contextual metadata. [[What are the challenges? Also talk about high-level design principles before delving into details in 4.1.]]

4.1 MOTION-AWARE QUALITY SCORER (MQS)

Feature extraction. MQS combines spatial artifact cues with aligned motion features to model the perceptual degradation of transcoded videos. We represent each video as a sequence of one-second chunks. Let v_t denote the t-th chunk and I_t its associated anchor frame. [[What is an anchor frame?]] These inputs provide four complementary views of the video content. *(a) Frame.* To capture global semantic context, we downscale the anchor frame I_t from $H \times W$ to $H' \times W'$ pixels. A spatial encoder converts the resized frame into high-level spatial tokens $\mathbf{H}_t$ that describe the overall scene. *(b) Patches.* To preserve fine-grained transcoding artifacts that may be attenuated by resizing, we extract P patches of $h \times w$ native pixels from the same anchor frame, using a grid-based sampling strategy inspired by FAST-VQA (Wu et al., 2022). For each patch, the same spatial encoder produces patch tokens $\mathbf{A}_t$ that retain local spatial details. *(c) Clip.* To capture global temporal dynamics, we again resize the frames of v_t and process them with a motion encoder (Feichtenhofer et al., 2019). The resulting spatiotemporal features are pooled into a global motion token $\mathbf{T}_t$. *(d) Tubes.* To

[1] Since DeQA and most MLLM-based quality scorers are no-reference models, we restrict this analysis to transcodes from 1080p and 2K sources, which are assumed to be pristine. Under this assumption, DMOS values are treated as absolute quality scores for the transcoded videos.

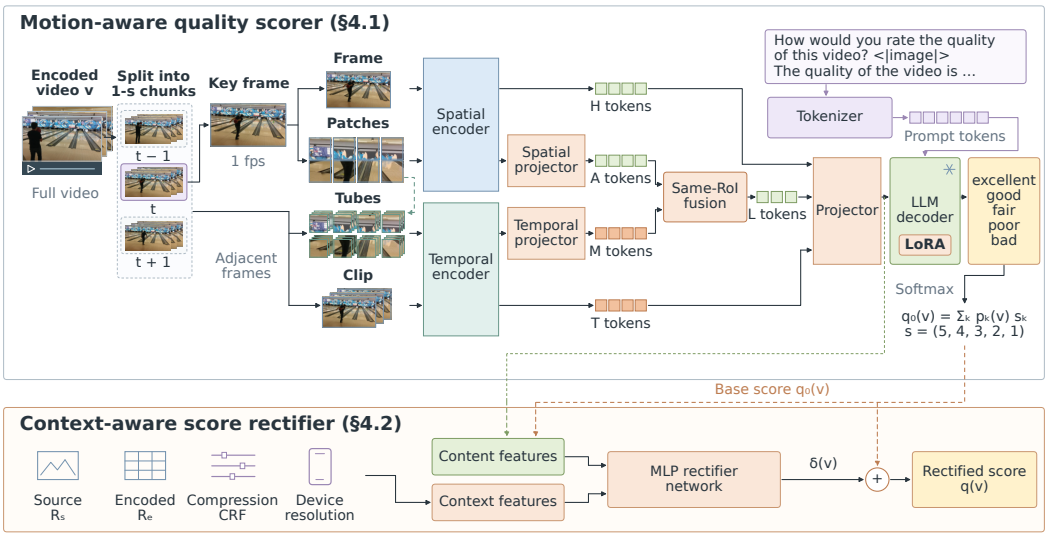


Figure 2: Overview of MoCA-VQA.

capture local motion, we extend each patch region over a short temporal neighborhood centered on I_t, forming a tube at a fixed spatial location. The same motion encoder produces local motion features $\mathbf{M}_t$ for these tubes. Each tube feature is spatially aligned with the corresponding patch tokens in $\mathbf{A}_t$.

We interleave patch and tube sampling across chunks by spatially offsetting their locations between consecutive chunks. Within each chunk, every patch remains aligned with its corresponding tube. This strategy expands spatial coverage across the video without increasing the number of patches or tubes per chunk.

Same-RoI fusion. We fuse each patch's artifact cues with motion features from the same region before aggregating across patches. This preserves their spatial correspondence and allows local artifacts to be interpreted in their motion context:

$$\mathbf{L}_t = \mathcal{F}_{\text{fuse}}(\mathbf{A}_t, \mathbf{M}_t). \tag{1}$$

The fusion function $\mathcal{F}_{\text{fuse}}$ applies learned feature interactions and gating to aligned patch and tube features, followed by pooling across patches and projection into the decoder's embedding space. The output $\mathbf{L}_t$ is a fixed-length sequence of motion-aware local tokens.

Previous MLLM-based VQA methods provide spatial tokens together with spatially pooled motion representations (Ge et al., 2024; Jia et al., 2024), which do not preserve explicit correspondences between local artifacts and motion. Retaining all local motion tokens would preserve more spatial detail but lengthen the decoder input and leave the relevant interactions to be learned implicitly. Even with positional information [cite], learning how motion modulates local distortion visibility may be difficult under limited video-level quality supervision. In contrast, MQS explicitly models these interactions before pooling, providing compact, motion-aware artifact representations at a comparable decoder token budget.

4.2 Context-aware Score Rectifier (CSR)

The source- and device-dependent effects in Sec. 3.2 motivate adapting of MLLM to be context-aware. To this end, we design a lightweight rectifier network, CSR and append it to MQS to compute the adjustment from video-derived *content features* and metadata-derived *context features*.

MLP rectifier network. We use the intermedia output of visual token from MQS and apply the principal component analysis (PCA) (Halko et al., 2011) to obtain a more compact video content feature. We concatenate this representation with a normalized version of the base score $q_0(v)$ to

form the content feature vector $\mathbf{c}_v$. We encode the source resolution, transcoding setting, and screen resolution into the context feature vector $\mathbf{k}_v$. [2] For video v under a given source, encoding, and display context, the rectified score is given by:

$$\delta(v) = \mathcal{F}_{\text{rect}}(\mathbf{c}_v, \mathbf{k}_v), \qquad q(v) = q_0(v) + \delta(v). \tag{2}$$

Here, $\mathcal{F}_{\text{rect}}$ includes the MLP network and its output constraints, and $\delta(v)$ denotes the correction under the specific context. For each video, the corrections are centered across screen tiers so that the mean prediction remains $q_0(v)$. Thus, the scorer determines the overall quality level, while the rectifier learns context-dependent deviations from that baseline.

Loss function. We supervise the predicted correction $\delta(v)$ using the difference between the video's DMOS in the given context and its mean DMOS across screen tiers. The training objective combines correction fitting, quality ranking, and regularization:

$$\mathcal{L}_{\text{CSR}} = \mathcal{L}_{\text{ctr}} + \lambda_r \mathcal{L}_{\text{rank}} + \lambda_\delta \mathcal{L}_\delta. \tag{3}$$

The fitting loss $\mathcal{L}_{\text{ctr}}$ trains the rectifier to predict the difference between each screen tier's DMOS and the video's mean DMOS across tiers. The ranking loss $\mathcal{L}_{\text{rank}}$ supervises the relative ordering of final predictions $q(v)$ across videos within each screen tier. The regularization term $\mathcal{L}_\delta$ penalizes large corrections to prevent excessive deviations from the base score. The coefficients λ_r and λ_δ control the contributions of ranking and regularization, respectively.

5 Experiments

5.1 Experimental Setup

Datasets and metrics. We evaluate the base scorer MQS on TSVQ and two public VQA datasets, KVQ (Lu et al., 2024) and KoNViD-1k (Hosu et al., 2017). We evaluate CSR on TSVQ. We report the Spearman rank-order correlation coefficient (SROCC), Kendall rank-order correlation coefficient (KROCC), Pearson linear correlation coefficient (PLCC), and root mean squared error (RMSE). Higher correlation coefficients and lower RMSE indicate better performance.

Implementation details. We initialize MQS from DeQA-Mix3 (You et al., 2025), a base model trained on multiple public image quality assessment (IQA) datasets. We fine-tune the LLM decoder using LoRA and jointly optimize the encoder and projector. Training runs for 3 epochs on 4 NVIDIA GPUs using AdamW (Loshchilov & Hutter, 2019), with an initial learning rate of 2×10^{-5}, cosine learning-rate decay (Loshchilov & Hutter, 2017), and an effective batch size of 128. Training takes approximately 6.5, 2.4, and 1.0 hours on TSVQ, KVQ, and KoNViD-1k, respectively. We train CSR separately with MQS frozen, using AdamW with a learning rate of 10^{-3}. The number of training epochs is selected based on validation performance. We retrain all baselines on the corresponding datasets following their respective training protocols. Our scorer and the MLLM-based baselines use language backbones of comparable size (7B–8B parameters). Additional implementation details are provided in Appendix **??**.

Baselines. For base quality prediction, we compare against seven MLLM-based methods: Q-Align (Wu et al., 2024), LMM-VQA (Ge et al., 2024), CP-LLM (Wen et al., 2025), VQA^2 (Jia et al., 2024), Q-Insight (Li et al., 2025), VQ-Insight (Zhang et al., 2026), and DeQA (You et al., 2025). We also include FAST-VQA (Wu et al., 2022), a Transformer-based method that achieves the strongest performance among the conventional approaches (Appendix **??**). For context adaptation, we compare CSR against the device-aware adaptation method of Safonov & Vatolin (2026). This method maps base quality scores and viewing conditions to context-specific predictions through a lightweight adaptation network, supervised by pairwise preferences derived from a context-conditioned Blade–Chest model.

[2] We obtain source information from metadata typically available in video delivery systems, without processing the source video itself.

Table 2: Comparisons of baseline MLLM-based VQA with MoCA-VQA. Brackets give 95% confidence intervals (CIs).

Method	Metric	TSVQ	KVQ	KoNViD-1k
Q-Align (Wu et al., 2024)	SROCC	0.646 [0.572, 0.724]	0.800 [0.773, 0.823]	0.831 [0.781, 0.869]
	PLCC	0.647 [0.580, 0.718]	0.807 [0.785, 0.829]	0.858 [0.823, 0.889]
FastVQA (Wu et al., 2022)	SROCC	0.794 [0.752, 0.835]	0.842 [0.820, 0.862]	**0.881** [0.843, 0.909]
	PLCC	0.809 [0.772, 0.847]	0.847 [0.828, 0.865]	**0.895** [0.868, 0.919]
LMM-VQA (Ge et al., 2024)	SROCC	0.050 [−0.053, 0.144]	0.442 [0.383, 0.497]	0.540 [0.436, 0.630]
	PLCC	0.064 [0.026, 0.194]	0.568 [0.521, 0.615]	0.619 [0.532, 0.701]
CP-LLM (Wen et al., 2025)	SROCC	0.472 [0.357, 0.593]	0.815 [0.788, 0.838]	0.809 [0.752, 0.855]
	PLCC	0.454 [0.357, 0.574]	0.822 [0.799, 0.843]	0.837 [0.794, 0.877]
VQA^2 (Jia et al., 2024)	SROCC	0.747 [0.676, 0.819]	0.853 [0.829, 0.873]	0.881 [0.842, 0.908]
	PLCC	0.726 [0.663, 0.791]	0.867 [0.851, 0.882]	0.878 [0.848, 0.905]
Q-Insight (Li et al., 2025)	SROCC	–	0.884 [0.867, 0.898]	0.861 [0.818, 0.894]
	PLCC	–	0.890 [0.877, 0.903]	0.878 [0.845, 0.906]
VQ-Insight (Zhang et al., 2026)	SROCC	–	0.652 [0.608, 0.691]	0.856 [0.813, 0.888]
	PLCC	–	0.686 [0.649, 0.722]	0.870 [0.840, 0.897]
DeQA (You et al., 2025)	SROCC	0.850 [0.806, 0.889]	0.904 [0.889, 0.917]	0.842 [0.794, 0.878]
	PLCC	0.867 [0.826, 0.907]	0.907 [0.895, 0.919]	0.860 [0.823, 0.892]
MQS (ours)	SROCC	**0.884** [0.847, 0.916]	**0.913** [0.899, 0.924]	0.858 [0.814, 0.891]
	PLCC	**0.896** [0.867, 0.924]	**0.915** [0.904, 0.925]	0.874 [0.845, 0.902]

Table 3: Fusion ablation on TSVQ.

Input views	Fusion approaches	SROCC	PLCC	RMSE	SROCC (between content)
Frame	–	0.850	0.867	0.258	0.478
Patches	–	0.801	0.797	0.314	0.465
Frame + Patches	–	0.871	0.888	0.238	0.555
Frame + Patches + Clip	–	0.876	0.889	0.238	0.538
All four	Token concatenation	0.868	0.883	0.243	0.519
All four	Local + global fusion	0.868	0.884	0.243	0.484
All four	Same-RoI cross-attention	0.877	0.893	0.234	0.529
All four	Same-RoI fusion (MQS)	**0.884**	**0.896**	**0.231**	**0.591**

5.2 Results

MLLM scorer. Table 2 compares MQS with existing no-reference VQA methods. We have three observations. First, MQS achieves the highest SROCC and PLCC on both TSVQ and KVQ, two datasets focused on short-form UGC. Compared with DeQA, MQS improves SROCC both within source-specific transcoding ladders (0.936 to 0.952) and across source-level mean scores (0.478 to 0.591) on TSVQ. These gains suggest that the additional spatial and motion cues support quality comparisons across both transcoding configurations and source contents. On both datasets, the paired 95% confidence intervals for the SROCC and PLCC gains over the baselines lie entirely above zero, indicating statistically significant improvements. Second, on KoNViD-1k, MQS achieves a higher SROCC than DeQA (0.858 vs. 0.842) but a lower score than FAST-VQA and VQA^2 (both 0.881). However, the SROCC and PLCC differences from these baselines are not statistically significant, as all paired 95% confidence intervals include zero. The limited training data in KoNViD-1k may constrain model adaptation. Third, video-oriented MLLMs such as LMM-VQA and VQA^2 do not consistently outperform frame-based scorers such as Q-Align and DeQA. This suggests that incorporating motion features alone does not guarantee more accurate quality prediction. The benefit of temporal modeling may depend on how motion cues are aligned and integrated with local artifacts. Similarly, the evaluated RL-based methods do not consistently outperform SFT-based scorers under our evaluation protocol, despite the improvements reported in their original studies (Li et al., 2025; Zhang et al., 2026). These results indicate that RL-based quality reasoning does not necessarily translate into improved scoring accuracy for transcoded-specific VQA task.

Table 4: Screen-context SROCC with device adaptation.

Screen tier	720p			1080p			>1080p			Average		
Scorer	base	+Flex	+CSR	base	+Flex	+CSR	base	+Flex	+CSR	base	+Flex	+CSR
Q-Align	0.591	0.579	0.576	0.637	0.622	0.666	0.594	0.584	0.604	0.607	0.595	0.615
FastVQA	0.703	0.703	0.708	0.781	0.781	0.803	0.740	0.740	0.737	0.741	0.741	0.749
DeQA	0.717	0.717	0.732	0.818	0.818	0.830	0.778	0.778	0.783	0.771	0.771	0.781
VQA^2	0.672	0.672	0.662	0.722	0.722	0.752	0.696	0.696	0.703	0.697	0.697	0.706
CP-LLM	0.410	0.410	0.423	0.468	0.466	0.491	0.433	0.435	0.440	0.437	0.437	0.451
MQS (ours)	0.746	0.746	**0.762**	0.853	0.853	**0.864**	0.813	0.813	**0.815**	0.804	0.804	**0.814**

Table 5: Rectifier ablation on TSVQ.

	Average			Screen-tier SROCC		
Variant	SROCC	PLCC	KROCC	720p	1080p	>1080p
MQS (base)	0.804	0.818	0.622	0.746	0.853	0.813
Screen-tier prompt	0.810	0.826	0.628	0.763	0.854	0.813
Learned tier token	0.803	0.815	0.620	0.758	0.847	0.804
Tier–resolution lookup	0.810	0.824	0.627	**0.764**	0.854	0.813
CSR w/o source/CRF	0.807	0.824	0.624	0.762	0.850	0.809
MQS + CSR	**0.814**	**0.828**	**0.633**	0.762	**0.864**	**0.815**

Fusion ablation. Table 3 shows an ablation study of different input views and fusion approaches for MQS. For input views, adding native-resolution *Patches* to *Frame* raises SROCC from 0.850 to 0.871. However, using only *Patches*, without the high-level frame tokens, reduces SROCC to 0.801. This suggests that global scene context complements local artifact cues when adapting MLLMs for VQA. Adding the global *Clip* view provides a small gain to 0.876, consistent with the use of global temporal cues in VQA^2 (Jia et al., 2024) and LMM-VQA (Ge et al., 2024). On the other hand, when all four views are available, the fusion approach also matters. Specifically, we study different versions of fusions: *Token concatenation* directly adds local motion tokens to the LLM decoder without patch-tube fusion; *Local + global fusion* retains local fusion and merges global frame and clip features into shared tokens; and *Same-RoI cross-attention* uses a cross-attention module before decoding to let patch tokens attend to the temporal states of the corresponding tube. *Same-RoI fusion*, used in MQS, achieves the best results, reaching 0.884 SROCC. Keeping global semantic and motion cues separate may preserve their complementary information more effectively than *Local + global fusion*. Compared with *Same-RoI cross-attention*, our fusion uses pooled motion features rather than learning attention over individual time steps, which may simplify learning from video-level quality labels. These results support fusing aligned artifact and motion features before aggregation.

Context-aware rectifier. Table 4 compares CSR with the SOTA device-aware adapter (Safonov & Vatolin, 2026) using the same frozen base predictions on TSVQ. CSR improves average SROCC for all six scorers, including an increase from 0.804 to 0.814 for MQS. In contrast, Flex leaves average SROCC nearly unchanged for most scorers. Its limited gains may reflect its content-agnostic adaptation: under the same device conditions, videos with identical base scores receive identical predictions, regardless of differences in their content. By incorporating pooled video content features, CSR learns content-dependent corrections, allowing such videos to receive different adjustments and potentially improving their ranking within each screen tier.

Rectifier ablation. Table 5 compares the variants of CSR. *Screen-tier prompt* retrains the scorer with the screen resolution tier specified in the prompt, while *Learned tier token* prepends a trainable screen-tier embedding to the visual token sequence. The remaining variants keep MQS frozen. *Tier-resolution lookup* adds a bias for each combination of screen tier and transcoded resolution which is regressed from training labels. *CSR w/o source/CRF* retains the full rectifier architecture and training objective but removes the source-resolution, transcoded-resolution, and CRF inputs. The complete

model, *MQS + CSR*, uses all content and context features described in Sec. 4.2. The full rectifier achieves the highest average SROCC (0.814) among all variants. Its gains over *Screen-tier prompt* and *Learned tier token* suggest that explicit score rectification is more effective than introducing screen information through text prompts or learned embeddings alone. The improvement over *Tier-resolution lookup* further supports learning content-specific corrections rather than applying shared biases based solely on screen tier and transcoded resolution. Finally, the performance drop without source and transcoding metadata indicates that these inputs provide useful information beyond video content features and screen tier.

6 Conclusion

The DMOS of a short-form transcode depends on the file, its upload and the screen tier (and its rater pool) it is judged on; single-file scorers model only the first. TSVQ measures it with the device logged per rating (12,600 transcodes, 377K valid ratings, 25 phones in three tiers) and separates what a scorer must perceive, temporal masking under heavy compression, from what it must be told: source-dependent quality differences and resolution gains that vary across screens. The MQS component of MoCA-VQA relates local motion to local artifacts on one patch grid and raises held-out TSVQ SROCC from 0.850 to 0.884 (0.913 on KVQ); CSR, which combines frozen content features with source, encoding and display metadata, raises tier-wise macro SROCC on the 709 matched videos from 0.804 to 0.814, most on 720p screens ($+0.016$). Logging the device per rating turned a nuisance variable into a label dimension that future subjective studies should record rather than control.

Limitations. Screen tier coincides with rater pool, so tier effects are associations; the 720p screen tier has 3 device codes, 4 raters and test labels averaging 2.37 ratings (96 of 709 singletons), and the modelling split holds only 1080p and 2K sources (Appendix **??**). The MQS results use a single training seed and the final checkpoint, the rectifier's PCA is fitted before its cross-validation folds and its epoch (117) sits near the 120 cap, and CSR inference additionally requires R_s and C, without which about 70% of the gain is forfeited.

AI use statement

We used generative AI tools for literature search, for drafting and editing the text, and for writing analysis and plotting code. The authors reviewed and revised all AI-drafted text, checked literature suggestions against the cited papers, and verified AI-written code against the underlying data. Generative AI produced none of the ratings, videos or trained models. We take responsibility for the final content of this work. [TODO: confirm categories and wording against the ICLR 2027 AI Policy for Authors]

Ethics statement

TSVQ is a subjective study with raters from a vendor annotation team, screened for vision and for age and sex distribution per ITU-R BT.500 (ITU-R, 2019). Released ratings carry an anonymous rater identifier and no personal data [TODO: confirm rater compensation, consent and ethics-review status]. Stimuli are uploads to a leading short-video platform [TODO: confirm content licensing and the release form of source and transcoded videos].

Reproducibility statement

We will release the DMOS labels with rating standard deviations, the per-tier labels and device-to-tier mapping, the split files, the training and evaluation code (MQS adaptation, CSR fitting) and checkpoints [TODO: confirm release scope, licence and hosting]. Section 3 describes the protocol, BT.500/P.910 processing and split; Section 4.1 the architecture, losses and optimisation of MQS; Section 4.2 the rectifier's inputs, objective and epoch selection; Section 5 the metric protocol. Run manifests (checkpoint ids, inference runs, prediction files) for Tables 2–5 will accompany the code [TODO: appendix table].

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